Date: 08/04/2026

EX-14
Log Analysis for Detection and Response

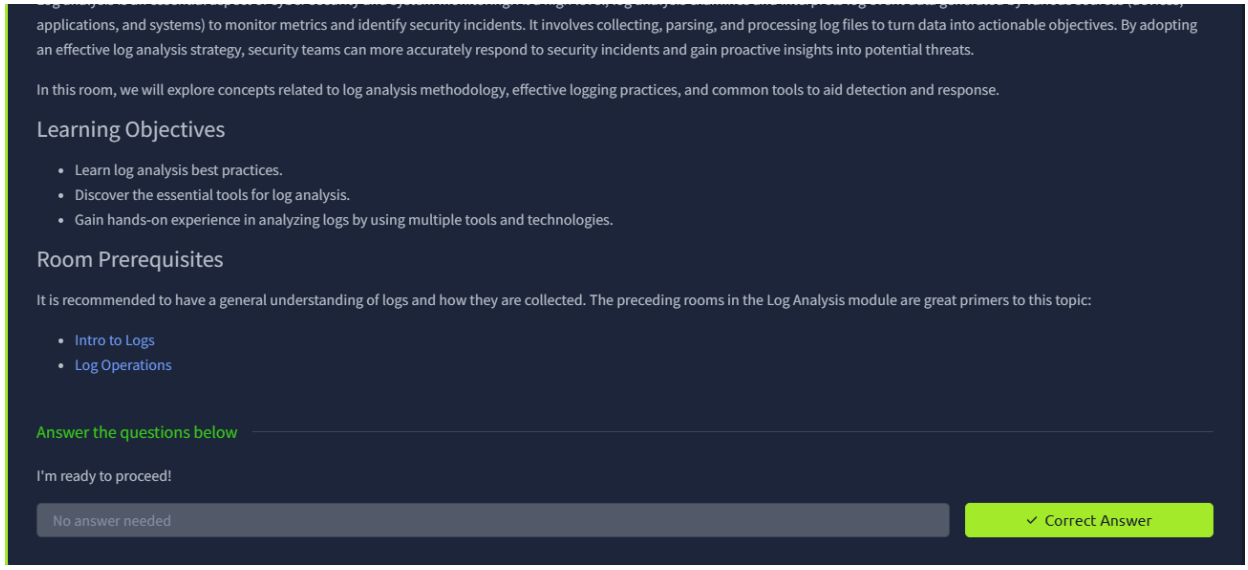
AIM:

To monitor and analyze system and network logs to detect suspicious activities, identify security incidents, and respond effectively to potential threats.

PROCEDURE:

- Collect logs from systems, servers, and network devices.
- Store and centralize logs using a SIEM (Security Information and Event Management) tool.
- Analyze logs for unusual patterns, anomalies, or indicators of compromise.
- Correlate events across multiple log sources to identify potential threats.
- Generate alerts for suspicious activities detected in the logs.
- Take appropriate response actions to investigate and mitigate the identified threat.

TASK-1:



applications, and systems) to monitor metrics and identify security incidents. It involves collecting, parsing, and processing log files to turn data into actionable objectives. By adopting an effective log analysis strategy, security teams can more accurately respond to security incidents and gain proactive insights into potential threats.

In this room, we will explore concepts related to log analysis methodology, effective logging practices, and common tools to aid detection and response.

Learning Objectives

- Learn log analysis best practices.
- Discover the essential tools for log analysis.
- Gain hands-on experience in analyzing logs by using multiple tools and technologies.

Room Prerequisites

It is recommended to have a general understanding of logs and how they are collected. The preceding rooms in the Log Analysis module are great primers to this topic:

- Intro to Logs
- Log Operations

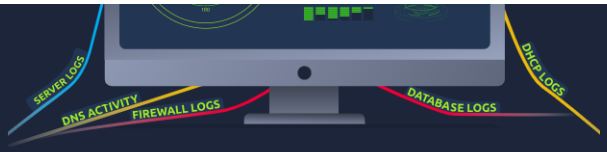
Answer the questions below

I'm ready to proceed!

No answer needed

✓ Correct Answer

TASK-2:



As discussed in the Intro to Logs room, different components within a computing environment generate various types of logs, each serving a distinct purpose. These log types include, but are not limited to:

- **Application Logs:** Messages from specific applications, providing insights into their status, errors, warnings, and other operational details.
- **Audit Logs:** Events, actions, and changes occurring within a system or application, providing a history of user activities and system behavior.
- **Security Logs:** Security-related events like logins, permission alterations, firewall activities, and other actions impacting system security.
- **Server Logs:** System logs, event logs, error logs, and access logs, each offering distinct information about server operations.
- **System Logs:** Kernel activities, system errors, boot sequences, and hardware status, aiding in diagnosing system issues.
- **Network Logs:** Communication and activity within a network, capturing information about events, connections, and data transfers.
- **Database Logs:** Activities within a database system, such as queries performed, actions, and updates.
- **Web Server Logs:** Requests processed by web servers, including URLs, source IP addresses, request types, response codes, and more.

Each log type presents a unique perspective on the activities within an environment, and analyzing these logs in context to one another is crucial for effective cyber security investigation and threat detection.

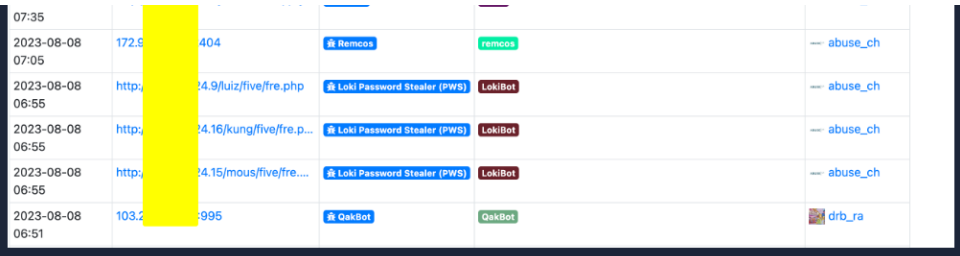
Answer the questions below

I understand the basics of logs and I'm ready to proceed!

No answer needed

✓ Correct Answer

TASK-3:



07:35				
2023-08-08 07:05	172.9 404	Remcos	remcos	abuse_ch
2023-08-08 06:55	http:/ 24.9/luiz/five/fre.php	Loki Password Stealer (PWS)	LokiBot	abuse_ch
2023-08-08 06:55	http:/ 24.16/kung/five/fre.p...	Loki Password Stealer (PWS)	LokiBot	abuse_ch
2023-08-08 06:55	http:/ 24.15/mous/five/fre....	Loki Password Stealer (PWS)	LokiBot	abuse_ch
2023-08-08 06:51	103.2 995	QakBot	QakBot	drb_ra

Using GREP to search a logfile for an IP address

```
cmnatic@thm grep "54.36.149.64" logfile.txt
54.36.149.64
```

Answer the questions below

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline

✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe` be classified as?

File Hashes

✓ Correct Answer

TASK-4:

`/etc/shadow`). A useful directory traversal payload list to reference can be found here.

It is important to note, like with the above examples, that directory traversals are often URL encoded (or double URL encoded) to avoid detection by firewalls or monitoring tools. Because of this, `%2E` and `%2F` are useful URL-encoded characters to know as they refer to the `.` and `/` respectively.

In the below example, a directory traversal attempt can be identified by the repeated sequence of `../` characters, indicating that the attacker is attempting to "back out" of the web directory and access the sensitive `/etc/passwd` file on the server.

path-traversal.log

```
10.10.113.45 - - [2023-08-05 18:17:25] "GET /../../../../../etc/passwd HTTP/1.1" 200 505
```

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log — ✓ Correct Answer

A log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Path Traversal — ✓ Correct Answer

TASK-5:

It is cheap and does not require expensive tooling. For example, simple Linux commands can do the trick.	It is time-consuming as the analyst has to do all of the work, including reformatting log files.
Allows for a thorough investigation.	N/A
Reduces the risk of overfitting or false positives on alerts from automated tools.	Events or alerts can be missed! Especially if there is a lot of data to comb through.
Allows for contextual analysis. The analyst has a broader understanding of the organization and cyber security landscape.	N/A

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated — ✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual — ✓ Correct Answer

TASK-6:

solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, the command line remains essential for quick and straightforward log analysis tasks.

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 — ✓ Correct Answer

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 — ✓ Correct Answer

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 — ✓ Correct Answer

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000 — ✓ Correct Answer

TASK-7:

In the configuration above, we use our previously defined regular expression pattern to extract IPv4 addresses from the "message" field of incoming log events. The extracted values will be added under the custom "ipv4_addresses" field name we defined. Typically, IP addresses are extracted automatically by default configurations. But this simple example shows the power of regular expression patterns when dealing with complex log files and custom field requirements.

The Logstash room and the official Grok documentation are fantastic resources for further exploring Logstash input and filter configurations!

Answer the questions below

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

post=2[0-9] | ✓ Correct Answer

What is the name of the filter plugin used in Logstash to parse unstructured log data?

Grok | ✓ Correct Answer

TASK-8:

Answer the questions below

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed | ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 | ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF_WIZARD} | ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 | ✓ Correct Answer

TASK-9:

Answer the questions below

What languages does Sigma use?

YAML | ✓ Correct Answer

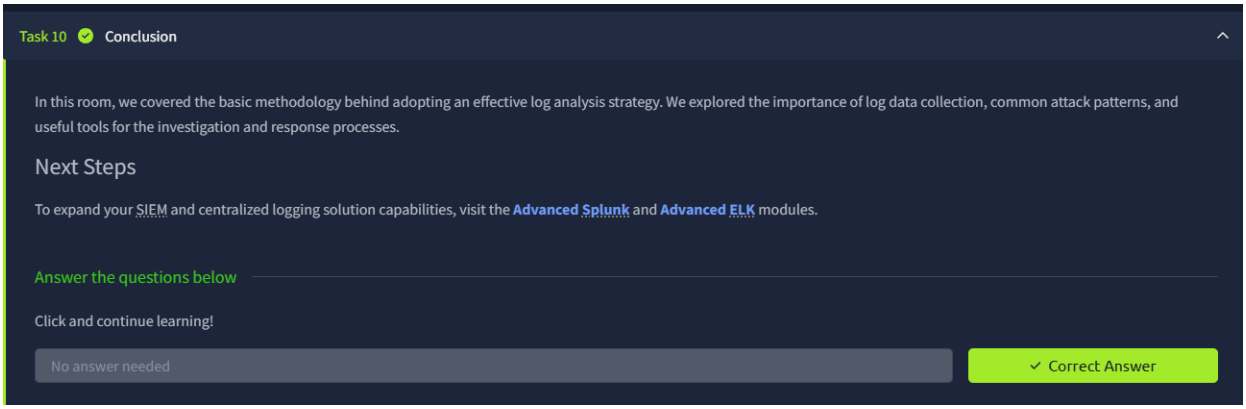
What keyword is used to denote the "title" of a Sigma rule?

title | ✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule | ✓ Correct Answer

TASK-10:



RESULT:

Log analysis was successfully performed across system, network, and web server logs using tools such as grep, CyberChef, Logstash, and ELK Stack. Suspicious activities including path traversal attacks, encoded payloads, and malicious IP connections were identified. The experiment demonstrated improved ability to detect security incidents from log data and enabled faster, more effective threat response, thereby reducing potential damage to the system.